

1411. Number of Ways to Paint N × 3 Grid

Difficulty: Hard

Link: <https://leetcode.com/problems/number-of-ways-to-paint-n-3-grid/?envType=daily-question&envId=2026-01-03>

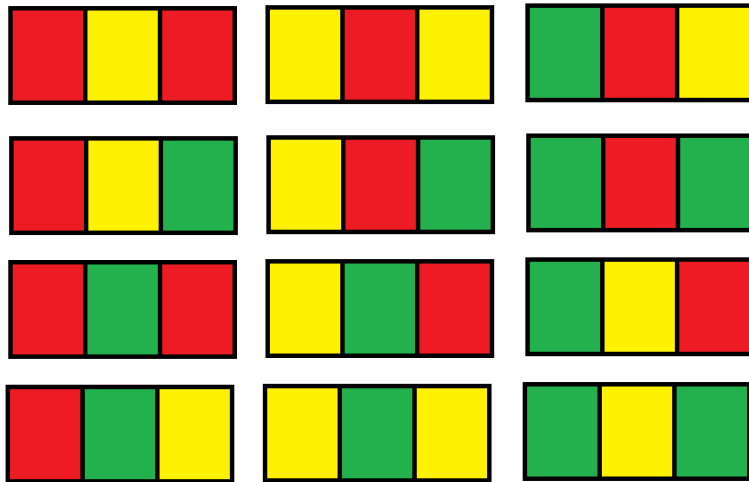
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Problem:

You have a grid of size $n \times 3$ and you want to paint each cell of the grid with exactly one of the three colors: **Red**, **Yellow**, or **Green** while making sure that no two adjacent cells have the same color (i.e., no two cells that share vertical or horizontal sides have the same color).

Given n the number of rows of the grid, return *the number of ways* you can paint this grid. As the answer may grow large, the answer **must be** computed modulo $10^9 + 7$.

Example 1:



Input: `n = 1`

Output: `12`

Explanation: There are 12 possible way to paint the grid as shown.

Example 2:

Input: `n = 5000`

Output: `30228214`

Constraints:

- `n == grid.length`
- `1 <= n <= 5000`